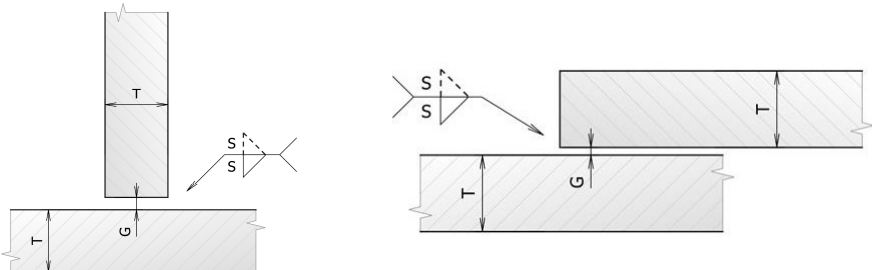
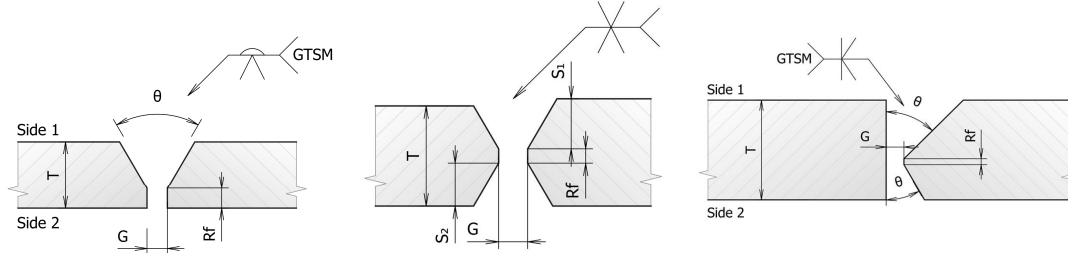
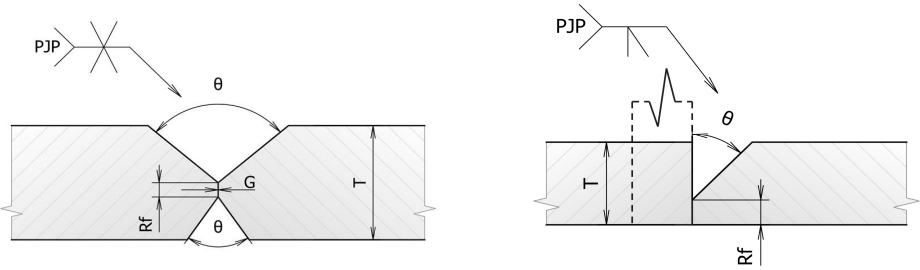
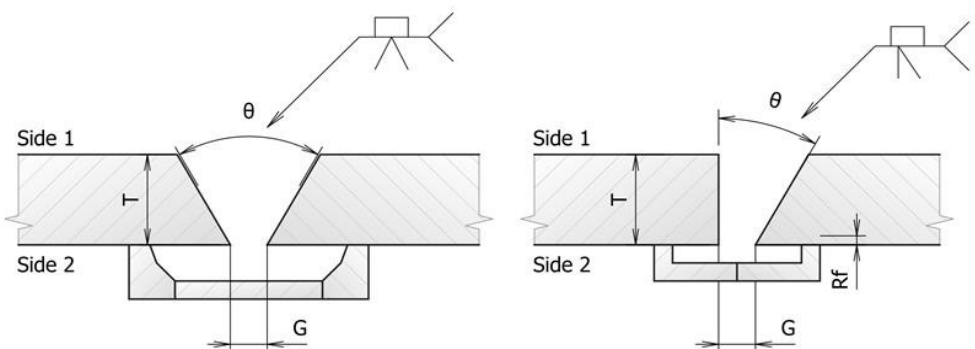


POLAR POCKET WPS						Pocket-POLAR-01-M		0	March 18, 2026		
						WPS No.		Rev.	Date		
 Seaspan Vancouver Shipyards Co. Ltd.						Applicable Standard(s)		Lloyds Register - Rules for the Manufacture, Testing and Certification of			
Process/Mode		Electrode (Wire) Classification				Brand Name(s)		Manufacturer(s)			
FCAW/Semi-Auto(Hand) FCAW/Auto(Moggy)		AWS A5.20 E71T-1M/9M-J LR Grade: 4Y40S				Dual Shield Prime 71 LT H4		ESAB			
Material Designation	Base material 1		Base material 2		Min. Preheat / Interpass Temp.	5°C for 3 ≤ T ≤ 20mm 10°C for 20 < T ≤ 40mm 65°C for 40mm < T ≤ 60mm 110°C for T > 60 (See Note 7, Note 8 & Note 9)					
	EH36, DH36 and all lower grades excluding A,B, D and E (Note 4 & Note 5)		EH36, DH36 and all lower grades excluding A,B, D and E (Note 4 & Note 5)								
Thickness or Dia	3 to 100 mm		3 to 100 mm		Max. Interpass Temp.	180°C					
Nominal Pipe Size	500mm and above		500mm and above		PWHT	N/A					
Welding Position	* All positions excluding Vertical down * Ceramic/Steel Backing: 1G, 2G and 3G-up ONLY.				Joint Design	CJP & PJP Groove Single or Double Bevel / Vee, T- K- Y Joint, Fillet					
TYPICAL JOINT CONFIGURATIONS ON NEXT PAGE											
COMPLETE JOINT PENETRATION		Welding Layer		JOINT TYPE		Back Purge		N/A		Contact Tip to Work Distance	9.5 - 20 mm (Target 15 mm)
Back-gouged to sound metal Welded onto backing Welded from one side without backing Welded both sides w/o back-gouging		Single/Multi Layer		BUTT		Backing type		If used, Ceramic/Steel		Interpass Cleaning	Grinding and Wire Wheel
		One/Two side		CORNER		Welding Technique		Stringer/Slight Weave		Shielding Gas	75% Argon / 25% CO2
		Gun travel angle ³		LAP		Max. Bead Width		Multi-Run: 18 mm Single-Run: 15 mm		Gas Flow	16-25 LPM
		Pull w/ slight push on Vu		TEE		No. of electrodes		1			34-53 CFH
Method of steel preparation		Oxy fuel/Plasma cut Grinding Milling		EDGE							
BM Thickness, T(mm)	Layers / Passes	Electrode Size (mm)	Welding Process	Power Mode	Consumable	Amperage (A)	Voltage (V)	WFS (IPM)	Travel Speed (mm/min)	Heat Input ¹ kJ/mm	
3 ≤ T ≤ 24	Single-Pass Fillet	1.2 - 1.6	FCAW (Semi-Auto)	DCEP	E71T-1M/9M	180 - 300	22 - 30	170 - 500	147 - 196	1.1 - 2.2	
			FCAW (Auto-Moggy)	DCEP	E71T-1M/9M	200 - 350	24 - 32	170 - 500	210 - 288	1.0 - 2.4	
3 ≤ T ≤ 24	Root Hot/Fill/Cap Multi-Run Fillet	1.2 - 1.6	FCAW (Semi-Auto)	DCEP	E71T-1M/9M	180 - 250	22 – 28	170 - 300	145 - 158	1.5 - 2.9	
			FCAW (Semi-Auto)	DCEP	E71T-1M/9M	180 - 350	22 - 29	170 - 500	382 - 474	0.5 - 1.6	
			FCAW (Auto-Moggy)	DCEP	E71T-1M/9M	180 - 350	22 - 29	170 - 500	382 - 474	0.5 - 1.6	
24 < T ≤ 100	Root Hot/Fill/Cap Multi-Run Fillet	1.2 - 1.6	FCAW (Semi-Auto)	DCEP	E71T-1M/9M	180 - 280	22 - 30	170 - 300	129 - 140	1.7 - 3.9	
			FCAW (Semi-Auto)	DCEP	E71T-1M/9M	150 - 350	20 - 30	170 - 500	242 - 360	0.5 - 2.6	
			FCAW (Auto-Moggy)	DCEP	E71T-1M/9M	180 - 400	22 - 30	170 - 500	185 - 339	0.7 - 3.9	
Note 1: Heat Input (kJ/mm) = [V x A x 60] / [Travel Speed (mm/min) x 1000]											
Note 2: Clean joint and adjacent areas to bright metal. Remove paint, primer, rust, scale, moisture, and contaminants. Wire brush between passes.											
Note 3: Travel angle 5–10° pull. For vertical up, use slight push.											
Note 4: Qualified for all delivery conditions except quenched and tempered steel.											
Note 5: Welding Grade A, B, D, E hull steel with Dual Shield Prime 71 LT requires Lloyd's special agreement for LR Class vessels.											
Note 6: For fillet welds, if root gap G > 3 mm, increase fillet leg size by (G – 2 mm). If G > 5 mm, consult Welding Engineering.											
Note 7: See Preheat & Interpass WI (SQMS-VSY-WEQ-P002) for other base materials.											
Note 8: If ambient temperature is < 10°C or moisture is present, preheat the joint to a minimum of 20°C prior to welding.											
Note 9: For ceramic backing up to 40 mm thickness, the recommended minimum preheat temperature for root and hot passes is 50°C.											
Note 10: For the root pass on the ceramic backing, it is recommended to use 1.2mm (0.045") wire size with max. 180 amps.											
Note 11: For ceramic backing, rectangular groove tiles recommended. If round ceramic is used, follow with GTSM.											
Note 12: If more information from the welding procedure is required, please refer to Weldeye or consult Welding Engineering.											
								Reference WPS No.			
Bug-O Weave Settings	Dwell (L&R): 0 - 1.0			Weave Width: 0-18				FC-CS-G-03 (Rev. 4) FC-CS-F-03 (Rev. 0)			
	Weave Pattern No.: #1 for 2G / #1 & #3 for other			Weave Speed: 0-50							

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Applicable Standard(s)				
TYPICAL JOINT CONFIGURATIONS				
Fillet Weld				
Fillet Leg Size (Multi-Run) = 4.2 to 140 mm				
Fillet Leg Size (Single-Run) = 5 to 12 mm				
Fillet Root Gap = 0 to 2 mm (See Note 6)				
Back-gouged to Sound Metal (GTSM)				
Root Gap (G) = 0mm to 3 mm for T up to 8 mm				
Root Gap (G) = 0mm to 4.5 mm for T over 8 mm				
Root Face (Rf) = 3mm minimum				
Groove angle (theta) = 40-70°				
Partial Joint Penetration (PJP)				
Root Gap (G) = 0 - 1mm				
Root Face (Rf) = 2mm minimum				
Groove angle (theta) = 40-70°				
Welded Onto Backing				
Root Gap for Ceramic (G) = Target 4 to 7 mm (max 1.5xT or 11mm, whichever smaller, is acceptable)				
Root Gap for Steel (G) = 5 mm minimum				
Root Face for Ceramic/Steel (Rf) = 0 to 1 mm				
Groove angle (theta) = 40 - 70°				
See Note 10, Note 11				